

M. Mahenoor

+91 81230 12684 | mahi090405@gmail.com | LinkedIn | GitHub | Ballari, Karnataka, India

Summary

Backend-focused software engineer who owns systems end-to-end — architecture, database design, API layer, and cloud deployment. Technical Lead on NEVARA, a geospatial intelligence platform with production AWS infrastructure serving paying NGO/CSR clients and working with real production data. Builds automated satellite-data pipelines, integrates third-party APIs, and ships Dockerized services independently in a fast-moving, resource-constrained startup environment.

Technical Skills

- **Languages:** Python (primary), SQL, C++, Bash
- **Backend & APIs:** Flask, FastAPI, REST API design, microservices, JWT/session-based auth, third-party API integration
- **Databases:** PostgreSQL, Supabase, schema design & query optimization, NoSQL concepts, AWS S3
- **Cloud & DevOps:** AWS (EC2, ECS, S3, Lambda) — Cloud Practitioner Certified, Docker, Linux, Git/GitHub
- **AI/ML & Data:** Pandas, NumPy, Scikit-learn, XGBoost, OpenCV, Google Earth Engine, SHAP/LIME interpretability
- **Visualization:** Power BI, Tableau (basics)
- **Other:** Geospatial pipelines

Projects

NEVARA — Geospatial Intelligence & Ecological Monitoring Platform Nov 2025 – Present
Technical Lead, Personal Startup Project | GitHub

- Architected and deployed production backend services in Python (Flask) on AWS (EC2, ECS, S3, Lambda); built the RESTful API layer that delivers geospatial environmental data to NGO and CSR clients on a PostgreSQL/Supabase datastore.
- Designed multi-tenant authentication and authorization plus database schemas and validation middleware, keeping environmental monitoring records accurate and access-controlled across client organizations.
- Built backend data pipelines ingesting multi-temporal Sentinel-2 and Landsat satellite imagery on real client data, cutting manual processing time by 70%.
- Containerized the platform with Docker for consistent, repeatable deployment across environments, supporting ecosystem partnerships and funding interest of up to INR 5 lakh.

BlueCarbon Ledger — Python, Flask, PostgreSQL Sep 2025 – Present
GitHub

- Designed a carbon-registry backend integrating satellite, sensor, and geospatial data via REST APIs, with a PostgreSQL schema built for MRV record storage.
- Built data ingestion pipelines with schema validation to keep multi-source environmental inputs traceable and verifiable.

AI-Powered Student Monitoring System — Python, Flask, OpenCV, SQL Jan 2025 – Present
GitHub

- Built a backend system with Flask REST APIs and SQL-backed automated logging around a real-time computer-vision pipeline (OpenCV, LBPH facial recognition), enabling scalable attendance tracking across multiple classrooms.

Urban Heat Island Prediction — Python, Google Earth Engine, XGBoost, SQL 2025 – Present
GitHub

- Built an end-to-end data pipeline with schema-consistent storage and structured preprocessing; tuned an XGBoost model to $R^2 = 0.70$ and $MAE = 1.19^\circ\text{C}$ on satellite-derived urban temperature data.

Education

Ballari Institute of Technology and Management (BITM) Ballari, India
B.Tech, Computer Science & Engineering (Data Science) | CGPA: 9.18/10.0 2023 – 2027

Achievements

- **AWS Cloud Practitioner Certified** — production cloud deployment, EC2/ECS/S3/Lambda proficiency.
- **Patent Filed** — decentralized system for carbon MRV and automated verification using multi-source geospatial and sensor data (under review).
- **1st Place, National Startup Hackathon, Mangaluru (2026)** — backend platform design & technical innovation, 500+ competing teams.
- **Top 10 Nationally, 36-Hour National Hackathon (2025)** — among 1,200+ participants.